Question 2 (Henry)

2026-10-09

Do students who have taken both 131 and 134 report greater interest in machine-learning-related fields than students who have taken 131 only?

Our second question surrounds the association between taking certain PSTAT courses and having specific areas of interest. Specifically, we want to see if taking PSTAT 131 and/or PSTAT 134 has any correlation in machine learning domain interests. We chose the 131/134 courses because its course description fits the topic of machine learning and its techniques pretty well. It is a course that is well known in the PSTAT department, and with the ongoing advancements in artificial intelligence and machine learning, we only feel that the latter is worth exploring through our survey data.

```
problem2 <- read_csv("~/Desktop/PSTAT197A/module0-class-survey-team9/data/processed/q2/problem2.csv")
```

```
## Rows: 49 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (11): prog.prof, math.prof, stat.prof, upper_div_courses, courses, domai...
## dbl  (4): response_id, prog.comf, math.comf, stat.comf
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dim(problem2)
```

```
## [1] 49 15
```

```
head(problem2)
```

```
## # A tibble: 6 x 15
##   response_id prog.prof prog.comf math.prof math.comf stat.prof stat.comf
##         <dbl> <chr>         <dbl> <chr>         <dbl> <chr>         <dbl>
## 1        1001 int               4 int               4 adv               5
## 2        1002 int               4 int               3 int               5
## 3        1003 beg               3 int               3 adv               4
## 4        1004 adv               4 adv               5 adv               5
## 5        1005 adv               4 int               4 adv               5
## 6        1006 int               4 int               2 adv               4
## # i 8 more variables: upper_div_courses <chr>, courses <chr>,
## #   domain_specialization <chr>, research <chr>, type <chr>, language <chr>,
## #   domains <chr>, areas <chr>
```

We have already joined the two datasets (**background-clean.csv** and **interest-clean.csv**) by `response_id`. We can see that there are 49 responders who consented to sharing both reponses. While there are 15 columns, we will mainly be using `courses` and `areas`.

Data Validation

We are assuming that students who have taken PSTAT 134 have also taken PSTAT 131 since it is a pre-requisite to that course. We checked this assumption by ensuring that every student who listed PSTAT 134 also included PSTAT 131.

```
course_check <- problem2 %>%
  mutate(
    took_131 = str_detect(courses, "\\bPSTAT 131\\b"),
    took_134 = str_detect(courses, "\\bPSTAT 134\\b")
  )

# Show everyone who took PSTAT 134 but not PSTAT 131
course_check %>%
  filter(took_134 & !took_131) %>%
  select(response_id, courses)
```

```
## # A tibble: 0 x 2
## # i 2 variables: response_id <dbl>, courses <chr>
```

Since there are no students who took PSTAT 134, but not PSTAT 131, our data is valid.

Now let's see the potential options for area interests:

```
possible_values <- unique(unlist(strsplit(as.character(problem2$areas), split = ",")))

possible_values
```

```
##  [1] "Spatial statistics or time series analysis"
##  [2] " Data acquisition and engineering"
##  [3] " Data visualization and interactive dashboards"
##  [4] " Predictive modeling"
##  [5] " generally"
##  [6] " Statistical models and inference"
##  [7] "Analysis or classification of images"
##  [8] " Spatial statistics or time series analysis"
##  [9] " Natural language processing and analysis of text"
## [10] " Model deployment and software or web integrations"
## [11] " Databases"
## [12] " Algorithms"
## [13] " Deep learning and neural networks"
## [14] "Deep learning and neural networks"
## [15] "Data acquisition and engineering"
## [16] "Model deployment and software or web integrations"
```

Out of the eleven options (not counting spacing discrepancies), we will define 'machine learning' interests as one of the following six options:

```
ml_areas <- c(
  "Analysis or classification of images",
  "Deep learning and neural networks",
  "Data acquisition and engineering",
  "Natural language processing and analysis of text",
  "Predictive modeling, generally",
  "Statistical models and inference, generally"
)
```

Again, this part is subjective since most of the options can be interpreted to fit into the scope of machine learning. However, we only chose six out of the eleven topics to provide an adequate representation of what is closest to machine learning.

PSTAT 131 vs non-PSTAT 131

To begin, we will explore students who have taken PSTAT 131 and students who have not taken PSTAT 131.

```
enrollment_table <- problem2 %>%
  mutate(
    course_group = if_else(
      str_detect(coalesce(courses, ""), "\\bPSTAT 131\\b"),
      "Took PSTAT 131",
      "Did not take PSTAT 131"
    )
  ) %>%
  count(course_group, name = "Number of Students") %>%
  mutate(
    Proportion = `Number of Students` / sum(`Number of Students`)
  )

save_table(enrollment_table, "pstat131_enrollment.csv")

kable(
  enrollment_table,
  col.names = c(
    "PSTAT 131 Enrollment",
    "Number of Students",
    "Proportion"
  ),
  digits = 3,
  caption = "Students by PSTAT 131 Enrollment"
)
```

Table 1: Students by PSTAT 131 Enrollment

PSTAT 131 Enrollment	Number of Students	Proportion
Did not take PSTAT 131	5	0.102
Took PSTAT 131	44	0.898

Here, we can see that only 5 students have not taken PSTAT 131, totaling around 10% of the respondents. On the other hand, 90% of students have taken PSTAT 131 at 44 students.

```r
ml_data <- problem2 %>%
  mutate(
    took_131 = str_detect(coalesce(courses, ""), "\\bPSTAT 131\\b"),
    took_134 = str_detect(coalesce(courses, ""), "\\bPSTAT 134\\b"),
    course_group = case_when(
      took_131 & took_134 ~ "Both courses",
      took_131 & !took_134 ~ "PSTAT 131 only",
      TRUE ~ NA_character_
    ),
    areas = coalesce(areas, "")
  ) %>%
  filter(!is.na(course_group)) %>%
  crossing(area = ml_areas) %>%
  mutate(selected = str_detect(areas, fixed(area))) %>%
  group_by(response_id, course_group) %>%
  summarise(
    n_ml_areas = sum(selected),
    .groups = "drop"
  )

ml_data
```

```
## # A tibble: 44 x 3
##    response_id course_group   n_ml_areas
##          <dbl> <chr>               <int>
##  1        1001 PSTAT 131 only          3
##  2        1003 PSTAT 131 only          6
##  3        1004 PSTAT 131 only          2
##  4        1005 PSTAT 131 only          6
##  5        1006 Both courses            3
##  6        1007 PSTAT 131 only          6
##  7        1008 PSTAT 131 only          5
##  8        1009 PSTAT 131 only          5
##  9        1010 Both courses            5
## 10        1012 PSTAT 131 only          3
## # i 34 more rows
```

Now we have added another component: the number of machine learning area interests each student selected. We have printed out the raw table to see this addition.

```r
ml_summary <- ml_data %>%
  group_by(course_group) %>%
  summarise(
    n = n(),
    mean_ml_areas = mean(n_ml_areas),
    median_ml_areas = median(n_ml_areas),
    .groups = "drop"
  )

save_table(ml_summary, "ml_interest_summary_by_pstat131.csv")

kable(
  ml_summary,
```

```
  digits = 3,
  caption = "ML-Related Interests by PSTAT 131 Enrollment"
)
```

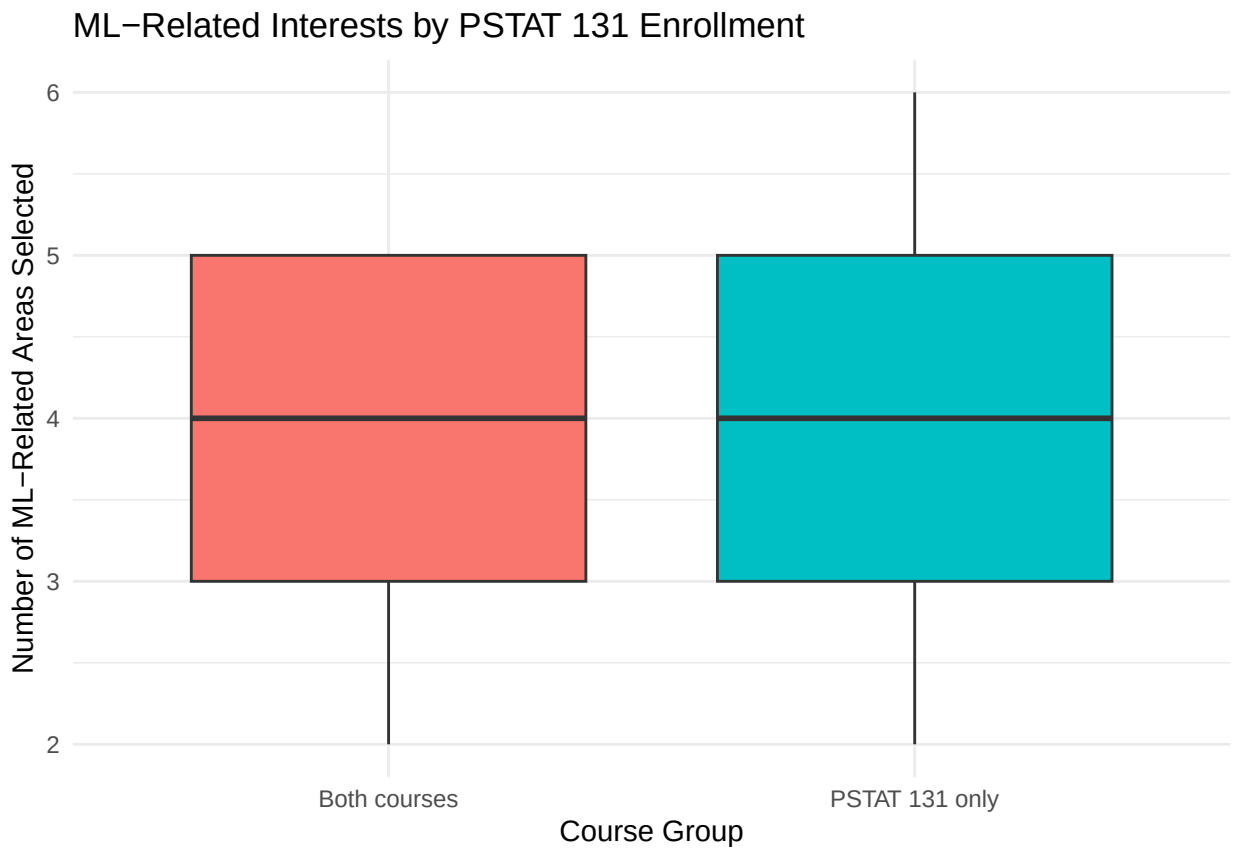
Table 2: ML-Related Interests by PSTAT 131 Enrollment

course_group	n	mean_ml_areas	median_ml_areas
Both courses	13	3.846	4
PSTAT 131 only	31	4.065	4

Here we have summary statistics for students who have taken PSTAT 131 vs students who have not taken PSTAT 131. The mean and median number of machine learning areas are shown. The mean is larger for students who took PSTAT 131, while the median is larger for students who have not taken PSTAT 131.

```
boxplot_131 <- ggplot(
  ml_data,
  aes(x = course_group, y = n_ml_areas, fill = course_group)
) +
  geom_boxplot() +
  labs(
    title = "ML-Related Interests by PSTAT 131 Enrollment",
    x = "Course Group",
    y = "Number of ML-Related Areas Selected"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

boxplot_131
```




```
ggsave(
  file.path(fig_dir, "fig_1_ml_interests_boxplot.png"),
  plot = boxplot_131,
  width = 8,
  height = 5,
  dpi = 300
)
```

Here we have a boxplot showing the distributions of each cohort. It appears that students who have not taken PSTAT 131 have a larger range, since there are only 5 students.

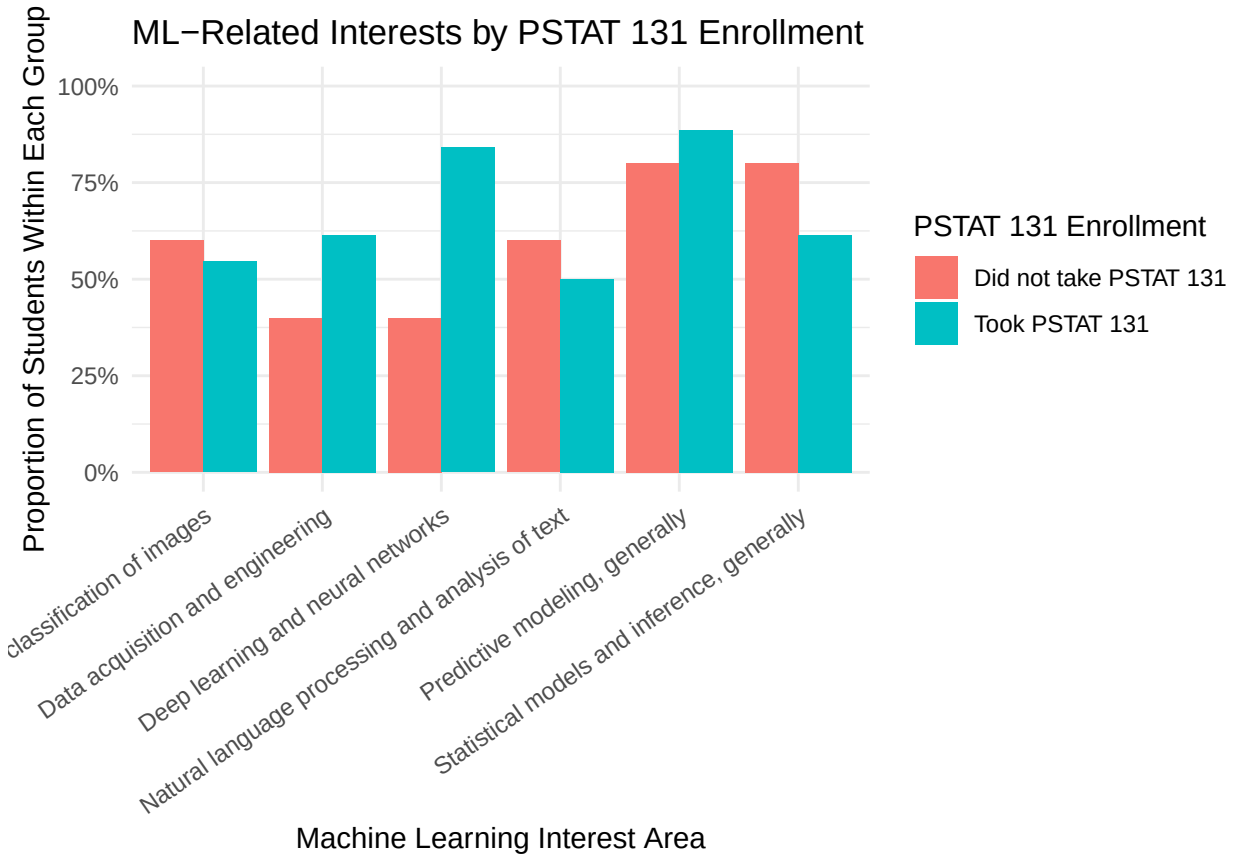
```
area_proportions <- problem2 %>%
  mutate(
    took_131 = str_detect(
      coalesce(courses, ""),
      "\\bPSTAT 131\\b"
    ),
    areas = coalesce(areas, "")
  ) %>%
  select(response_id, took_131, areas) %>%
  crossing(area = ml_areas) %>%
  mutate(
    selected = str_detect(areas, fixed(area)),
    course_group = if_else(
      took_131,
```

```
      "Took PSTAT 131",
      "Did not take PSTAT 131"
    )
  ) %>%
  group_by(course_group, area) %>%
  summarise(
    n_selected = sum(selected),
    n_students = n_distinct(response_id),
    proportion = n_selected / n_students,
    .groups = "drop"
  )

save_table(
  area_proportions,
  "ml_area_proportions_by_pstat131.csv"
)

area_chart <- ggplot(
  area_proportions,
  aes(x = area, y = proportion, fill = course_group)
) +
  geom_col(position = "dodge") +
  scale_y_continuous(
    labels = percent_format(),
    limits = c(0, 1)
  ) +
  labs(
    title = "ML-Related Interests by PSTAT 131 Enrollment",
    x = "Machine Learning Interest Area",
    y = "Proportion of Students Within Each Group",
    fill = "PSTAT 131 Enrollment"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 35,
      hjust = 1
    )
  )

area_chart
```




```
ggsave(
  file.path(fig_dir, "fig_2_ml_area_proportions.png"),
  plot = area_chart,
  width = 11,
  height = 7,
  dpi = 300
)
```

We can see that 'Deep learning and neural networks' has the biggest gap in proportion of students who find interest in that topic.

```
wilcox_result <- wilcox.test(
  n_ml_areas ~ course_group,
  data = ml_data,
  exact = FALSE
)

capture.output(
  wilcox_result,
  file = file.path(table_dir, "wilcoxon_test.txt")
)

wilcox_result
```

```
##
```

```
##  Wilcoxon rank sum test with continuity correction
##
## data:  n_ml_areas by course_group
## W = 186, p-value = 0.6887
## alternative hypothesis: true location shift is not equal to 0
```

We conducted a nonparametric test (particularly the Wilcoxon rank sum test) since the group of students who haven't taken PSTAT 131 is fairly small, at 5 students. This test is used to determine whether two independent samples come from populations with the same distribution or median. Since the p-value is high enough, we fail to reject the null hypothesis and conclude that there is no difference in the number of machine learning interests between students who have or have not taken PSTAT 131.

PSTAT 134 vs PSTAT 131

Now we will answer our initial question of whether taking PSTAT 131 or also taking PSTAT 134 has any correlation with the number of machine learning interests selected.

```
comparison_summary <- ml_data %>%
  group_by(course_group) %>%
  summarise(
    n = n(),
    mean_ml_areas = mean(n_ml_areas),
    median_ml_areas = median(n_ml_areas),
    .groups = "drop"
  )

save_table(
  comparison_summary,
  "ml_interest_summary_131_vs_131_134.csv"
)

kable(
  comparison_summary,
  digits = 3,
  caption = "ML Interests: PSTAT 131 Only vs Both Courses"
)
```

Table 3: ML Interests: PSTAT 131 Only vs Both Courses

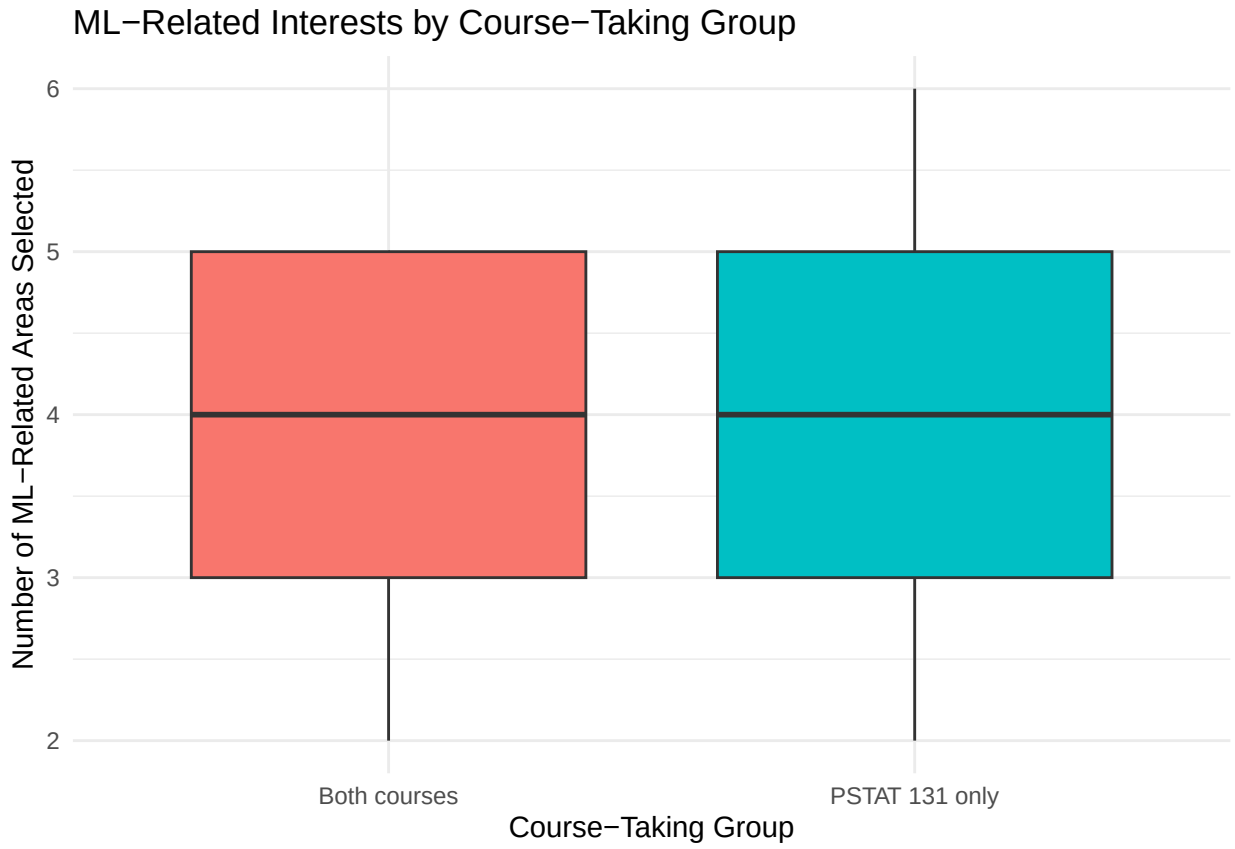
course_group	n	mean_ml_areas	median_ml_areas
Both courses	13	3.846	4
PSTAT 131 only	31	4.065	4

Here we can see that the mean number of machine learning interests is slightly higher for PSTAT 131 only students (4.065) compared to students who also took PSTAT 134 (3.846). The median, however, is the same for both groups at 4.

```
comparison_boxplot <- ggplot(
  ml_data,
  aes(x = course_group, y = n_ml_areas, fill = course_group)
) +
```

```
  geom_boxplot() +
  labs(
    title = "ML-Related Interests by Course-Taking Group",
    x = "Course-Taking Group",
    y = "Number of ML-Related Areas Selected"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

comparison_boxplot
```




```
ggsave(
  file.path(fig_dir, "fig_3_ml_interests_131_vs_131_134.png"),
  plot = comparison_boxplot,
  width = 8,
  height = 5,
  dpi = 300
)
```

The distributions for both groups seem to be almost the same, with the only exception being students taking PSTAT 134 lacking an upper tail.

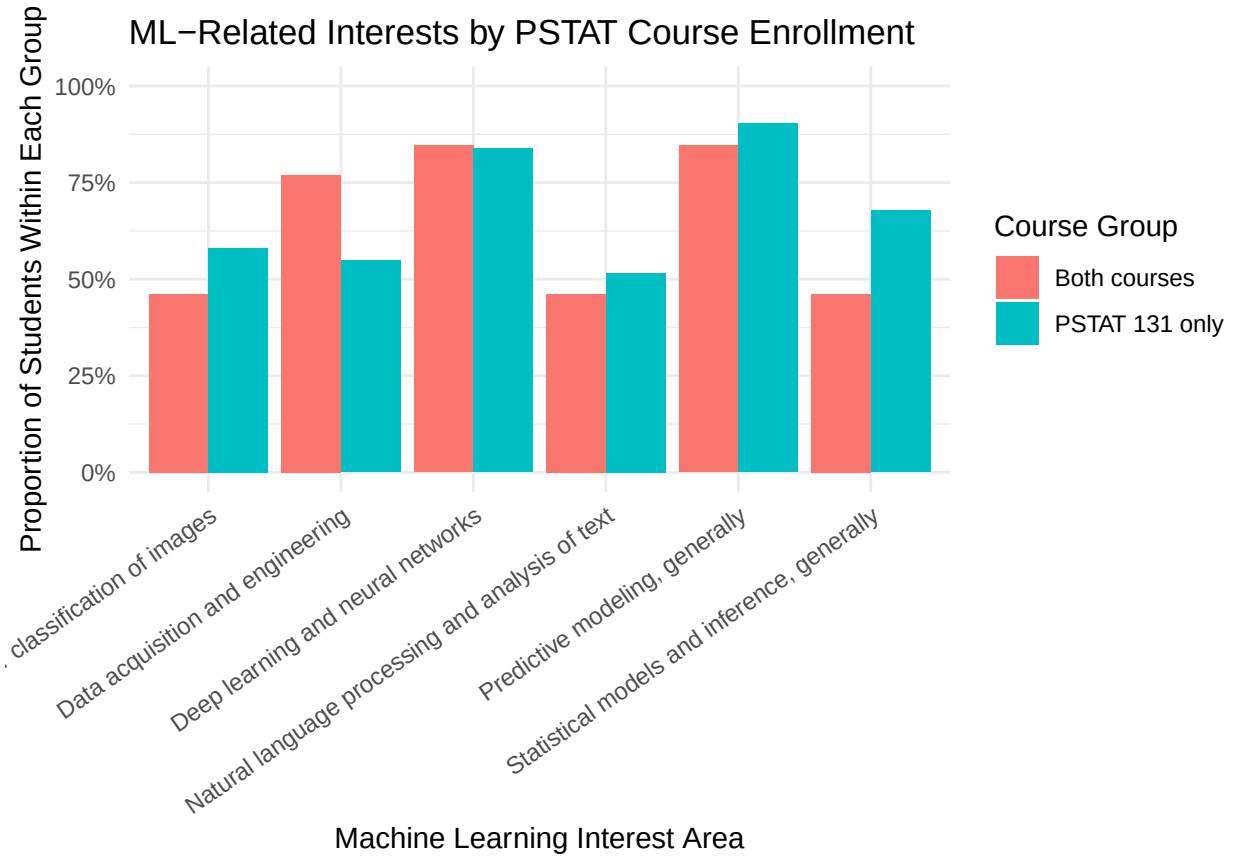
```
comparison_proportions <- problem2 %>%
  mutate(
    took_131 = str_detect(coalesce(courses, ""), "\\bPSTAT 131\\b"),
```

```
    took_134 = str_detect(coalesce(courses, ""), "\\bPSTAT 134\\b"),
    course_group = case_when(
      took_131 & took_134 ~ "Both courses",
      took_131 & !took_134 ~ "PSTAT 131 only",
      TRUE ~ NA_character_
    ),
    areas = coalesce(areas, "")
  ) %>%
  filter(!is.na(course_group)) %>%
  crossing(area = ml_areas) %>%
  mutate(selected = str_detect(areas, fixed(area))) %>%
  group_by(course_group, area) %>%
  summarise(
    n_selected = sum(selected),
    n_students = n_distinct(response_id),
    proportion = n_selected / n_students,
    .groups = "drop"
  )

save_table(
  comparison_proportions,
  "ml_area_proportions_131_vs_131_134.csv"
)

comparison_chart <- ggplot(
  comparison_proportions,
  aes(x = area, y = proportion, fill = course_group)
) +
  geom_col(position = "dodge") +
  scale_y_continuous(
    labels = scales::percent_format(),
    limits = c(0, 1)
  ) +
  labs(
    title = "ML-Related Interests by PSTAT Course Enrollment",
    x = "Machine Learning Interest Area",
    y = "Proportion of Students Within Each Group",
    fill = "Course Group"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 35,
      hjust = 1
    )
  )

comparison_chart
```




```
ggsave(
  file.path(fig_dir, "fig_4_ml_area_proportions_131_vs_131_134.png"),
  plot = comparison_chart,
  width = 11,
  height = 7,
  dpi = 300
)
```

Besides 'Data acquisition and engineering', it appears that students who only took PSTAT 131 have a higher probability of choosing the other 5 interests.

```
# Separate the two groups
both_courses <- ml_data %>%
  filter(course_group == "Both courses") %>%
  pull(n_ml_areas)

pstat131_only <- ml_data %>%
  filter(course_group == "PSTAT 131 only") %>%
  pull(n_ml_areas)

# One-sided Wilcoxon rank-sum test
test_result <- wilcox.test(
  both_courses,
  pstat131_only,
  alternative = "greater",
  exact = FALSE
```

```
)

test_result
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  both_courses and pstat131_only
## W = 186, p-value = 0.6654
## alternative hypothesis: true location shift is greater than 0
```

Again, we do not have convincing evidence that there is a difference in the number of machine learning interests between those who took PSTAT 131 and those who also took PSTAT 134.

Drawbacks

Some drawbacks about this research question is that the survey allowed the respondent as many choices they wanted for area interests. In other words, someone could have checked all the boxes and make the cohorts even harder to determine. Ideally, we would have each person only selected one topic they are most interested in.

Also, since there were multiple questions on the survey, with the interest question being near the end, many students may have felt fatigued and not thinking as deep for their true answers.